MSE603_Assignment 1

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Q1:

1. (20 points) A hospital employs volunteers to staff the reception desk between 8:00 AM and 10:00 PM. Each volunteer works four consecutive hours except for those starting at 8:00 PM who work for two hours only. The minimum need for volunteers is approximated by a step function over 2-hour intervals starting at 8:00 AM as 5, 7, 8, 6, 5, 6, and 8. Because most volunteers are retired individuals, they are willing to offer their services at any hour of the day (8:00 AM to 10:00 PM). However, because of the large number of charities competing for their service, the number needed must be kept as low as possible. Formulate an LP to determine the optimal schedule for volunteers.

Solution:

(1) Decision variables

(1) Define Variables:

x_i = the # of volunteers starting at hour i, $x_i \geq 0$

$i = 1, 2, 3 \cdots 13$ (8am, 10am, 12am, ... 8Pm)

(Each volunteer works 4 consecutive hours;
those starting at 8 Pm work 2 hours only.
So, no volunteer starts at 9Pm and 10 Pm)

(2) Objective function:

$$\min \sum_{i=1}^{13} x_i$$

s.t:

8:00 AM (hour 1)	$x_1 \geq 5$
9:00 AM (hour 2)	$x_1 + x_2 \geq 5$
10:00 AM (hour 3)	$x_1 + x_2 + x_3 \geq 7$
11:00 AM (hour 4)	$x_1 + x_2 + x_3 + x_4 \geq 7$
12:00 AM (hour 5)	$x_2 + x_3 + x_4 + x_5 \geq 8$
1:00 PM (hour 6)	$x_3 + x_4 + x_5 + x_6 \geq 8$
2:00 PM (hour 7)	$x_4 + x_5 + x_6 + x_7 \geq 6$
3:00 PM (hour 8)	$x_5 + x_6 + x_7 + x_8 \geq 6$
4:00 PM (hour 9)	$x_6 + x_7 + x_8 + x_9 \geq 5$
5:00 PM (hour 10)	$x_7 + x_8 + x_9 + x_{10} \geq 5$
6:00 PM (hour 11)	$x_8 + x_9 + x_{10} + x_{11} \geq 6$
7:00 PM (hour 12)	$x_9 + x_{10} + x_{11} + x_{12} \geq 6$
8:00 PM (hour 13)	$x_{10} + x_{11} + x_{12} + x_{13} \geq 8$
9:00 PM	$x_{11} + x_{12} + x_{13} \geq 8$

$x \geq 0, \forall i$

Q2:

2. (20 points) The demand of ice cream during the three summer months (June, July, and August) at All-Flavors Parlor is estimated at 310, 400, and 310 10-gallon cartons, respectively. Two wholesalers, 1 and 2, supply All-Flavors with its ice cream. Although the flavors from the two suppliers are different, they are interchangeable. The maximum number of cartons either supplier can provide is 400 per month. Also, the price the two suppliers charge change from one month to the next according to the following schedule:

	Price per carton in month		
	June	July	August
Supplier 1	\$100	\$110	\$120
Supplier 2	\$115	\$108	\$125

To take advantage of price fluctuation, All-Flavors can purchase more than is needed for a month and store the surplus to satisfy the demand in a later month. The cost of refrigerating an ice-cream carton is \$10 per month. It is realistic in the present situation to assume that the refrigeration cost is a function of the average number of cartons on hand during the month. Develop a model to determine the optimum schedule for buying ice cream from the two suppliers.

Solution:

① Decision variables.

- X_{ij} : the # of purchased cartons in month i from supplier j
  $i=1$, June ; $i=2$, July ; $i=3$, August
  $j=1$, Supplier 1 ; $j=2$, Supplier 2.
- S_i : the storage at the end of month i

② Objective function:

$$\min Z = 100X_{11} + 115X_{12} + 110X_{21} + 108X_{22} + 120X_{31} + 125X_{32} + 10 \times \sum_{i=1}^{3} \frac{(S_{i-1} + S_i)}{2}$$

③ Constrains: s.t. $X_{ij} \le 400$, $\forall i = 1,2,3$, $\forall j = 1,2$.

$X_{11} + X_{12} - S_1 = 310$

$X_{21} + X_{22} + S_1 - S_2 = 400$

$X_{31} + X_{32} + S_2 - S_3 = 310$

$X_{ij} \geq 0 \quad \forall i \ \forall j$, $S_i \geq 0$

Q3:

3. (20 points) A refinery manufactures two grades of jet fuel, F1 and F2, by blending four types of gasoline, A, B, C and D. Fuel F1 uses gasolines A, B, C, and D in the ratio 1:1:2:4, and fuel F2 uses the ratio 2:2:1:3. The supply limits for A, B, C, and D are 800, 1000, 1100, and 1400 bbl/day, respectively. The costs per bbl for gasolines A, B, C, and D are \$100, \$80, \$90, and \$140, respectively. Fuels F1 and F2 sell for \$220 and \$260 per bbl, respectively. The minimum demand for F1 and F2 is 240 and 430 bbl/day, respectively. Develop an LP model to determine the optimal production mix for F1 and F2.

Solution:

① Decision variables:

- X_{ij} : the # bbl of gasoline i in fuel j.
  $i = A, B, C, D$ types of gasoline.
  $j = 1, F_1$; $j = 2, F_2$.
- F_i : the # bbl of two grades of jet fuel
  $i = 1, F_1$; $i = 2, F_2$.

② Objective function.

$$\text{Max } z = 220F_1 + 260F_2 - 100(X_{A1} + X_{A2}) - 80(X_{B1} + X_{B2}) - 90(X_{C1} + X_{C2}) - 140(X_{D1} + X_{D2})$$

③ Constrains :

s.t. $F_1 \geq 240$, $F_2 \geq 430$

$X_{A1} + X_{A2} \leq 800$

$X_{B1} + X_{B2} \leq 1000$

$X_{C1} + X_{C2} \leq 1100$

$X_{D1} + X_{D2} \leq 1400$

$F_1 = X_{A1} + X_{B1} + X_{C1} + X_{D1}$, $X_{A1} = X_{B1}$, $X_{C1} = 2X_{A1}$, $X_{D1} = 4X_{A1}$

$F_2 = X_{A2} + X_{B2} + X_{C2} + X_{D2}$, $X_{A2} = X_{B2} = 2X_{C2}$, $X_{D2} = 3X_{C2}$

$X_{ij} \geq 0 \quad \forall ij$

Q4:

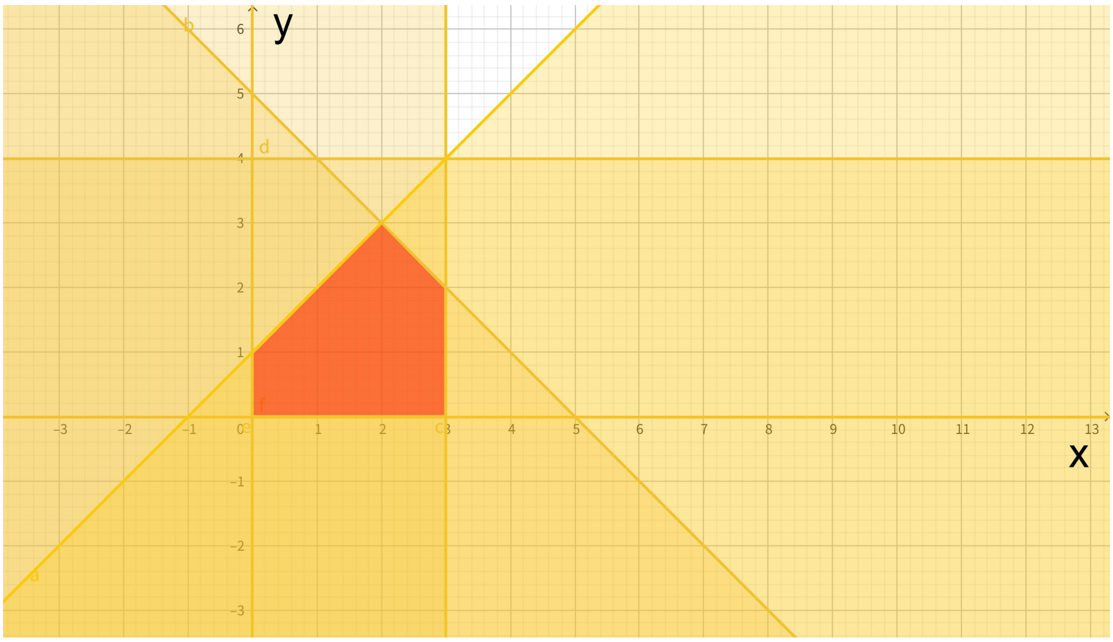
4.Consider the following linear program:

$$\max \; x_1 + 2x_2$$
$$\text{s.t.}$$
$$x_1 - x_2 \geq -1$$
$$x_1 + x_2 \leq 5$$
$$x_1 \leq 3$$
$$x_2 \leq 4$$
$$x_1, x_2 \geq 0$$

(a) (5 points) Graph the feasible region of the LP. Is the feasible region unbounded?
(b) (5 points) Are any of the above constraints redundant? If so, indicate which one(s).
(c) (10 points) Solve the LP using the graphical method. Explain your approach.
(d) (5 points) Is there more than one optimal solution? If so, give two different solutions. If not, explain using the graphical method why not?
(e) (15 points) Solve the LP using the Simplex method? Compare your solution with the one you got from part c). Are they the same?

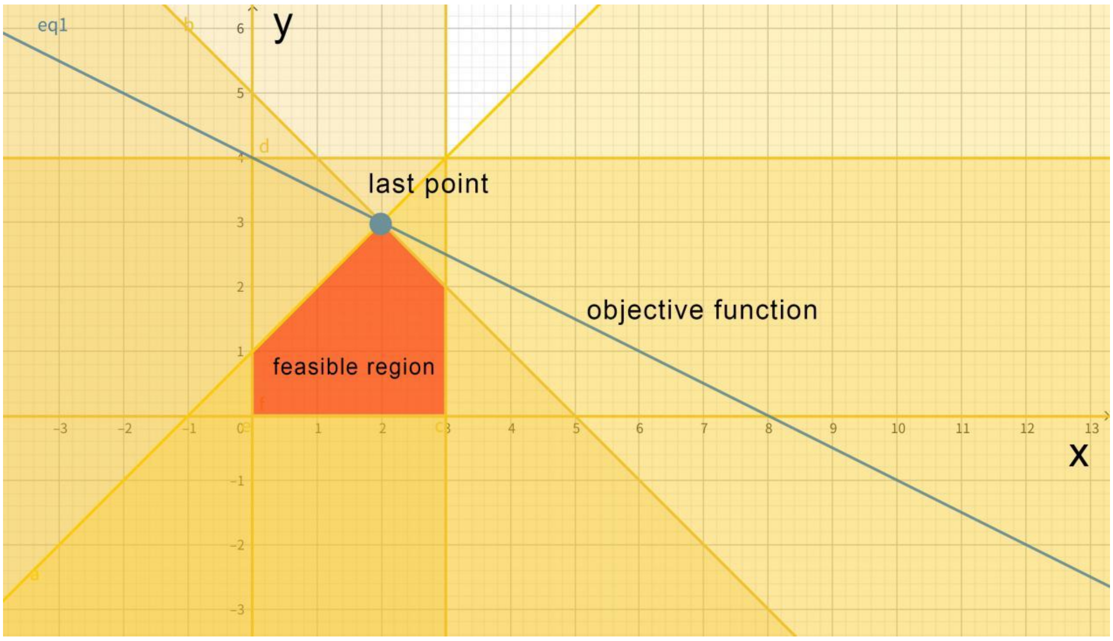
Solution:

(a) The feasible region of the LP. is bounded; it is shown in red in the graph.



(b) Yes, the constraint $x_2 \leqslant 4$ is redundant.

(c) Draw the objective function on the graph. Move this line until the last point of contact with the feasible region; the point is the optimal solution.
Optimal solution: values of decision variables at optimality are $x_1=2$, $x_2=3$.
Objective function value at optimality : $x_1+2\ x_2 = 8$.



(d) No. There is only one optimal solution because the objective line $x_1+2\ x_2$ is not parallel to any constraint line, so it touches the feasible region at a single extreme point.

(e) Solve the LP using the Simplex method:

① First, introduce the variable, z, and re-write the objective function

max z.

s.t.

$$
\begin{aligned}
z - x_1 - 2x_2 &= 0 \quad \text{Row 0}\\
-x_1 + x_2 + s_1 &= 1 \quad \text{Row 1}\\
x_1 + x_2 + s_2 &= 5 \quad \text{Row 2}\\
x_1 + s_3 &= 3 \quad \text{Row 3}\\
x_2 + s_4 &= 4 \quad \text{Row 4}
\end{aligned}
$$

$x_1, x_2, s_1, s_2, s_3, s_4 \geq 0$

② Then, we form the initial Simplex Tableau.

z	x_1	x_2	s_1	s_2	s_3	s_4	RHS	ratio
1	-1	$\boxed{-2}$	0	0	0	0	0	
0	-1	1	1	0	0	0	1	2/1=2
0	1	1	0	1	0	0	5	5/1=5
0	1	0	0	0	1	0	3	none
0	0	1	0	0	0	1	4	4/1=4

$BV = \{s_1, s_2, s_3, s_4\}$

$NBV = \{x_1, x_2\}$

③ From Row 1, we can know: $x_2 = 1 + x_1 - s_1$

Then, we transfer the Simplex Tableau.

z	x_1	x_2	s_1	s_2	s_3	s_4	RHS	ratio
1	$\boxed{-3}$	0	2	0	0	0	2	
0	-1	1	1	0	0	0	1	none
0	2	0	-1	1	0	0	4	4/2=2
0	1	0	0	0	1	0	3	3/1=3
0	1	0	-1	0	0	1	3	3/1=3

$BV = \{x_2, s_2, s_3, s_4\}$

$NBV = \{x_1, s_1\}$

④ From Row 2, we can know: $x_1 = 2 + \frac{1}{2}s_1 - \frac{1}{2}s_2$

Then, we transfer the Simplex Tableau.

z	x_1	x_2	s_1	s_2	s_3	s_4	RHS	ratio
1	0	0	0.5	1.5	0	0	6	
0	0	1	0.5	0.5	0	0	3	
0	1	0	-0.5	0.5	0	0	2	
0	0	0	0.5	-0.5	1	0	1	
0	0	0	-0.5	-0.5	0	1	1	

$BV = \{x_1, x_2, s_3, s_4\}$

$NBV = \{s_1, s_2\}$

⑤ Therefor, we can know the optimal solution is: $x_1 = 2$, $x_2 = 3$.

So, the answer is the same with the graph's result.